

Credit Card Customer & Transaction Analytics Dashboard (Power BI Project)

1. Project Name

Credit Card Customer & Transaction Analytics Dashboard

2. Problem Statement

- Credit card companies process millions of transactions and manage diverse customer segments.
- Without analytics systems it becomes difficult to understand spending behavior, revenue drivers, and product performance.
- This project analyzes credit card customer demographics, transactions, and card category performance.
- The objective is to help financial institutions identify growth opportunities, improve customer targeting, and optimize revenue strategies.

3. Techniques Used

- SQL used for database creation and importing credit card transaction datasets.
- Power Query used for data cleaning, transformation, and preparation.
- Star schema data modeling linking customer, card, and transaction datasets.
- DAX measures to calculate KPIs such as Total Revenue, Interest Earned, Transaction Count, and Customer Satisfaction Score.
- Interactive Power BI dashboards for customer segmentation, transaction trends, and card performance analysis.

4. Key Insights

- Total revenue reached \$57M with \$8M generated from interest income.
- The system processed \$46M in transactions across 667K transactions.
- Male customers spend 22–30% more than female customers, indicating a potential untapped female market.
- The 40–50 age group represents the largest and most valuable customer segment.
- The basic Blue card generates approximately 83% of total revenue, while premium cards underperform.
- Bills, fuel, and entertainment represent the largest spending categories.
- Revenue is highly concentrated in five states: Texas, New York, California, Florida, and New Jersey.

5. Business Recommendations

- Target female customers with tailored credit card benefits and marketing campaigns.
- Introduce student and young professional credit cards to capture the 20–30 age segment.
- Upsell high■spending customers to premium cards through upgrade incentives.
- Partner with utility companies, fuel stations, and grocery retailers for cashback programs.
- Expand marketing efforts to underserved states to reduce geographic revenue concentration.
- Improve customer satisfaction by enhancing service quality and mobile app experience.